Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

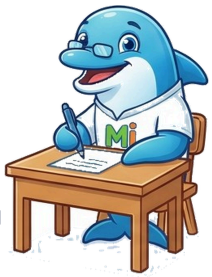
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big number puzzles	Not in this proof edition				☐
3	Operation detectives	Not in this proof edition				☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal detectives	Not in this proof edition				☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and Nets	6-7	8-9	10 and 12	11	☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

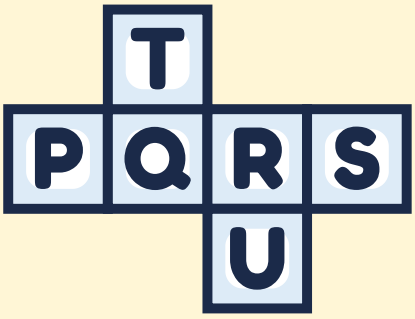
Learn it: Symmetry and nets

My goal. I can tell which nets fold into a cube and which faces are opposite. I can complete a figure so that a line is a line of symmetry.

A net is a flat shape that folds into a solid. A cube net has 6 squares. In a line of 3 squares, the 2 end squares are opposite faces. On a dice, opposite faces add up to 7.

Worked example 1

Does this net fold into a cube? Which face is opposite P?

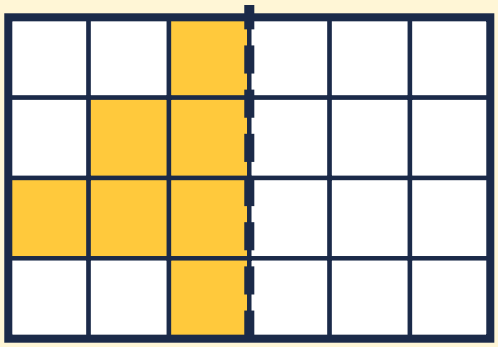


1. Choose a base. The row of 4 squares wraps round as walls.
2. One square is above the row and one is below. They make the lid and the base.
3. P and R have 1 square between them. They are opposite.

Answer: It folds into a cube. R is opposite P.

Worked example 2

Complete the figure. The dashed line must be a line of symmetry.



1. Flip each shaded square across the line.
2. The partner is the same distance from the line.

Answer: Each shaded square gets a partner the same distance from the line.

How to do it

1. Net: choose a base. Fold the squares round it in your head.
2. A net needs 1 base, 1 lid and 4 walls.
3. Symmetry: count squares from the line to the partner.

Good to know

- Dice: opposite faces add up to 7.
- A rectangle has 2 lines of symmetry. A square has 4.

Your turn

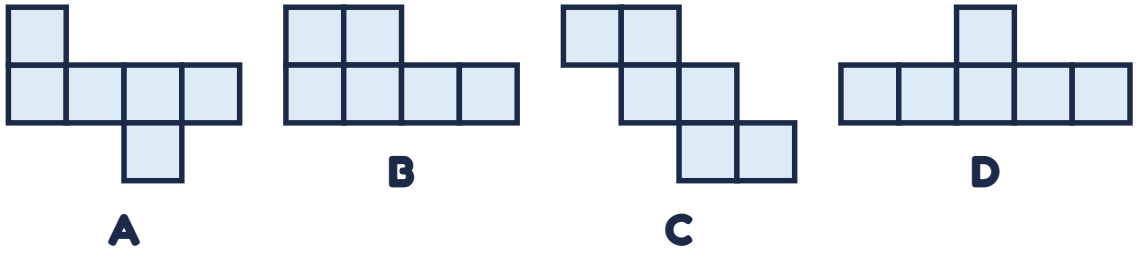
Answer at the back of the book.

Try **Opposite faces of a dice add up to 7. What number is opposite 4?** ☐

Start here

Do these first. Take your time.

1 **Which of these nets fold into a cube? Circle the letters.**



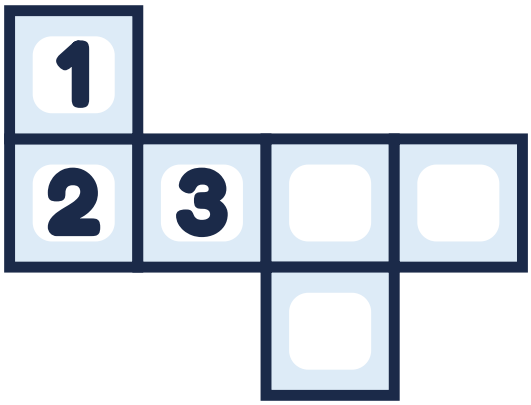
2 **The net folds into a cube. Name the letter opposite each one.**



(a) Opposite P: ☐ (b) Opposite Q: ☐

(c) Opposite S: ☐

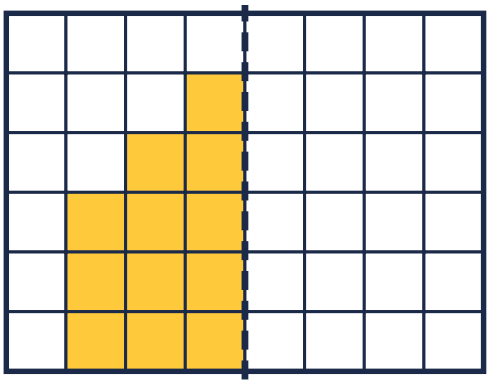
3 **Each pair of opposite faces on a dice has a total of 7. Fill in the blank squares. Go row by row.**



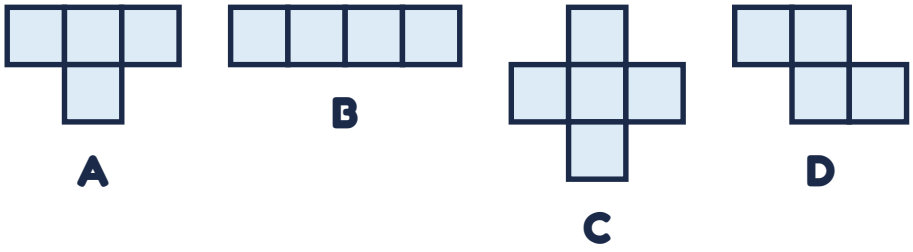
Start here

Keep going. Check your work.

4 Fold on the dashed line. Shade squares so the 2 halves match. How many squares are shaded in all? ☐



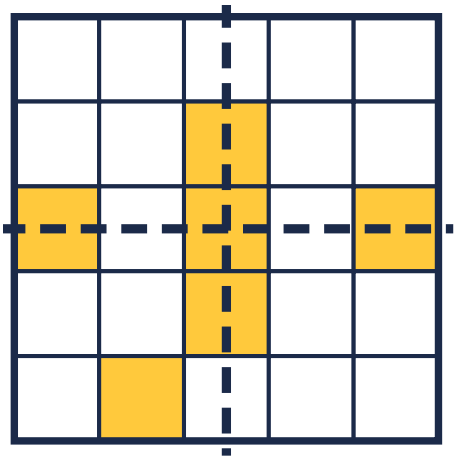
5 How many lines of symmetry does each shape have?



(a) A: ☐ (b) B: ☐ (c) C: ☐ (d) D: ☐

6 There are 2 dashed lines. Shade 1 more square. The halves must then match on one of the lines.

(a) How many different squares can you shade? ☐



Start here

Nearly there. You can do it.

7 **A dice sits on a table. The top face shows 6 and the face in front shows 4.**

(a) The bottom face shows ☐

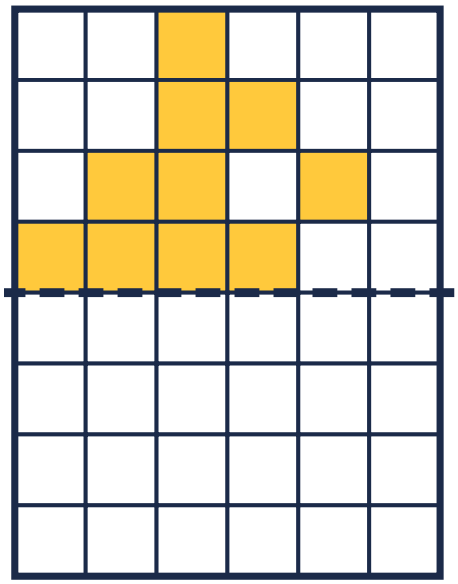
(b) The face at the back shows ☐

(c) The 4 side faces add up to ☐

8 **A box is 4 cm long, 3 cm wide and 2 cm high. Its net is cut from card. Find the area of the card.** ☐ **cm²**

Working: _____

9 **Shade squares so the top and bottom halves match on the dashed line. How many are shaded in all?** ☐



10 **A cube has edges of 5 cm.**

(a) All 12 edges are 5 cm long. The edges add up to ☐ cm

(b) The area of its net is ☐ cm²

Think harder

Hints are upside down. Try first!

1 **The faces of a cube are numbered 1 to 6. Face 1 touches 2, 3, 5 and 6. Face 2 touches 1, 3, 4 and 5.**

(a) Opposite 1 is ☐

(b) Opposite 2 is ☐

(c) Opposite 3 is ☐

Working: _____

Hint. A face touches 4 faces. It is opposite the 1 it does not touch.

2 **Mei sees 3 faces of a dice at the same corner. Their numbers add up to 10.**

(a) The numbers she can see, smallest first: ☐ ☐ ☐

(b) The numbers she cannot see, smallest first: ☐ ☐ ☐

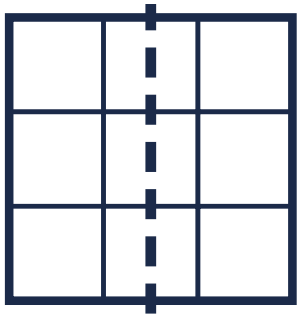
Working: _____

_____

Hint. At a corner, the 3 faces are from 3 different opposite pairs.

3

On a 3 by 3 grid, shade exactly 3 squares. A fold down the middle must make the halves match. How many patterns can you make? ☐



List: _____

_____

_____

Hint. Middle-column squares match themselves. The others come in pairs.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 4: Painted cubes**

Paint all the outside faces of a big cube that is 3 by 3 by 3. Then cut it into 27 small cubes.

(a) Small cubes with 3 painted faces: ☐

(b) Small cubes with 2 painted faces: ☐

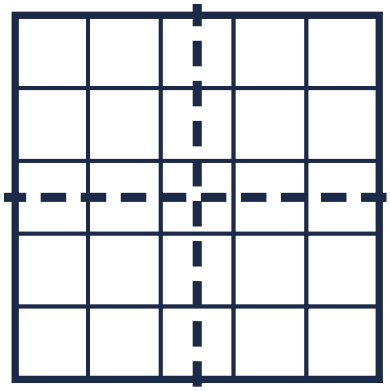
(c) Small cubes with 1 painted face: ☐

(d) Small cubes with no paint: ☐

Hint. Corners, edges and faces of the big cube. Count each kind.

2

Shade exactly 5 squares of a 5 by 5 grid. The pattern must have 4 lines of symmetry. How many different patterns are there? ☐



Hint. The middle square is always shaded. The other 4 squares go together.

3 **A dice has 1 on top, 2 facing north and 3 facing east. It rolls north, then east, then north again. A roll north puts the north face on the table. A roll east puts the east face on the table.**

(a) The top face is now ☐

(b) The bottom face is now ☐

Top, north, east: _____

_____

_____

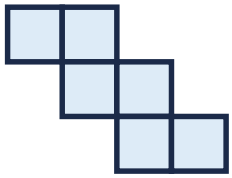
Hint. Write the top, north and east faces after each roll.

✔ Check yourself

No hints. Try each one on your own.

1

Does this net fold into a cube? **Yes** **No**



2 **A rectangle has a different length and width. How many lines of symmetry does it have?** ☐

Working: _____

3 **A dice has 2 on top and 3 in front. The bottom shows** ☐ **and the back shows** ☐ **.**

Working: _____

4 **A cube has edges of 4 cm. Its net has an area of** ☐ **cm².**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and Nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 8 · Symmetry and Nets

Your turn
page 6

Answer: 3
Why: 4 + 3 = 7, so 3 is opposite 4.

START HERE

1
page 7

Answer: A and C
Why: Fold each net in your head. A: the row of 4 makes the walls, and one square above and one below make the lid and the base. It folds. B: the 2 by 2 block of squares has 4 squares meeting at a corner, but only 3 faces of a cube meet at a corner. D: 5 squares in a row wrap round and the 5th covers the 1st. C: the staircase folds into a cube.

2
page 7

Answer: (a) R **(b)** T **(c)** U
Why: In the row of 3 squares P, Q, R, the 2 ends P and R are opposite. Q and T are opposite (1 square between them going round the step). The last pair is S and U.

3
page 7

Answer: 5, 4, 6 (row by row)
Why: Find the opposite squares. 2 and the third square of the row are opposite, so that square is 7 − 2 = 5. 3 and the fourth square are opposite, so it is 7 − 3 = 4. 1 and the bottom square are opposite, so it is 7 − 1 = 6.

4
page 8

Answer: 24
Why: There are 12 shaded squares on the left of the line. Each needs a partner on the right, so 12 more squares are shaded. 12 + 12 = 24.

5
page 8

Answer: (a) 1 **(b)** 2 **(c)** 4 **(d)** 0
Why: A has 1 line (down the middle). B has 2 lines (along its length and across its middle). C, the plus sign, has 4 lines (up-down, across and the 2 diagonals). D has none: it looks the same when turned half way round, but no fold makes the halves match.

6
page 8

Answer: (a) 2
Why: Look for a shaded square with no partner. Only the one in the bottom row has none. For the up-down line, its partner is in the bottom row on the right. For the across line, its partner is in the top row, above it. Each fix works for one line, so there are 2 ways.

7
page 9

Answer: (a) 1 **(b)** 3 **(c)** 14
Why: Opposite faces add up to 7. The bottom is 7 − 6 = 1. The back is 7 − 4 = 3. The 4 side faces are 2 opposite pairs, and 7 + 7 = 14.

8
page 9

Answer: 52
Why: The 6 faces are 2 of 4 by 3 (12 each), 2 of 4 by 2 (8 each) and 2 of 3 by 2 (6 each). 24 + 16 + 12 = 52 cm².

9
page 9

Answer: 20
Why: There are 10 shaded squares above the line. Each needs a partner below the line, so 10 more squares are shaded. 10 + 10 = 20.

10
page 9

Answer: (a) 60 (b) 150
Why: (a) 12 × 5 = 60 cm. (b) Each face is 5 × 5 = 25 cm², and the net has 6 faces: 6 × 25 = 150 cm².

THINK HARDER

1
page 10

Answer: (a) 4 (b) 6 (c) 5
Why: Face 1 does not touch 4, so 4 is opposite 1. Face 2 does not touch 6, so 6 is opposite 2. The faces left over are 3 and 5, so they are opposite.

2
page 10

Answer: (a) 1, 4, 5 (b) 2, 3, 6
Why: The pairs are 1 and 6, 2 and 5, 3 and 4. She sees one number from each pair, so no two of the 3 numbers add up to 7. The only choice with a total of 10 is 1, 5 and 4. The hidden faces are opposite: 6, 2 and 3.

3
page 10

Answer: 10
Why: The 3 shaded squares are 3 from the middle column, or 1 from the middle column and 1 pair. 3 in the middle column: 1 way. 1 in the middle column (3 choices) and 1 pair (3 rows to choose): 3 × 3 = 9 ways. 1 + 9 = 10.

GO FURTHER

1
page 11

Answer: (a) 8 (b) 12 (c) 6 (d) 1
Why: 3 painted faces: the 8 corner cubes. 2 painted faces: 1 cube in the middle of each of the 12 edges, so 12. 1 painted face: 1 cube in the middle of each of the 6 faces, so 6. No paint: the cube in the very middle, so 1. Check: 8 + 12 + 6 + 1 = 27.

2
page 11

Answer: 4
Why: With 4 lines of symmetry, the squares come in groups. The middle square is 1 group. The other groups have 4 squares each. They are the 4 corners, the 4 squares next to the corners on the diagonals, the 4 middle squares of the sides and the 4 squares touching the centre. One of these groups and the middle square make 5 squares: 4 patterns.

3
page 11

Answer: (a) 6 (b) 1
Why: Start: top 1, bottom 6, north 2, south 5, east 3, west 4. Roll north: the north face 2 lands on the table, so the top is 5, the bottom is 2, the north face is 1 and the south face is 6. Roll east: the east face 3 lands on the table, so the top is 4, the bottom is 3, the east face is 5 and the west face is 2. Roll north: the north face 1 lands on the table, so the bottom is 1 and the top is the old south face, 6.

CHECK YOURSELF

1
page 12

Answer: Yes
Why: The staircase of 2, 2, 2 squares folds into a cube. Each step wraps round to make the base, walls and lid with no overlaps.

2
page 12

Answer: 2
Why: It can be folded along the middle in 2 ways, and the halves match. The diagonals do not work.

3
page 12

Answer: bottom 5, back 4
Why: Opposite faces add up to 7. The bottom is 7 − 2 = 5 and the back is 7 − 3 = 4.

4
page 12

Answer: 96
Why: Each face is 4 × 4 = 16 cm². The net has 6 faces: 6 × 16 = 96 cm².